

**ACADEMIC YEAR: 2022-2023**

The main working principle of the DAB is based on the phase shift between the full bridges. The full bridge in the input will generate the square wave voltage V_1 on the transformer primary winding. Similarly, the full bridge in the output would generate the voltage V_2 on the secondary winding of the transformer. The square wave voltages have a phase shift with respect to each other caused by the full bridges. This phase shift is responsible for transferring the power from one side to the other

one. In general, the power transfers from the side with the leading voltage to the other side. [2]

If the power semiconductor devices and the transformer are assumed as ideal, it's possible to draw the equivalent circuit of the converter as in Figure 2.

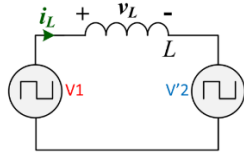


Figure 2 equivalent circuit of DAB converter [2]

where V_2' is nV_2 and "n" is the transformer windings turn ratio. L is the series inductance of the circuit including the leakage inductance of the transformer.

One of the main advantages of DAB converter is the ZVS. There are various ways to achieve ZVS in general. In DAB the ZVS is achieved by using the anti-parallel body diode of the switches. In other words, if the current flows through the switch's diode at the moment of the switching, then the voltage on the switch would be zero and hence the switch can be turned on at zero voltage. Therefore, it can be said that negative current at the moment of switching (with respect to the switch direction) is the sufficient condition for achieving ZVS-on in DAB. On the other hand, the ZVS can be lost for power lower than nominal power. Based on the modulation method, the ZVS range of DAB can change. [2]

The total power losses in DAB can be estimated as:

$$P_{DAB-loss} = P_{Tx} + P_L + P_{cap} + P_{SW} \quad \text{Eq. 1}$$

where P_{Tx} represents the total losses in the transformer, P_L indicates the total losses in the inductor, P_{cap} the total losses in the capacitors and P_{sw} the total losses in the switches [3]

Because of the fast switching in the DC-DC converters, some nodes have a great dv/dt . This results in the generation of displacement currents through the parasitic capacitances of the elements and ground, which lead to common-mode (CM) noises. This CM voltages (CMV) can flow through the grid via the cables and disturb other devices and cause electromagnetic interference. Moreover, they can cause the fault tripping of some protection devices. The CMV are affected by many parameters such as switching speed, operating frequency, and power level. In order to investigate the CMV in DAB all the parasitic capacitances in the circuit should be identified. The parasitic

capacitances in DAB are consists of the ones from the switches to ground via the heatsink, the ones in the transformer and general parasitic capacitances such as the ones between the DC links and the ground. Considering all the capacitances, the model in Figure 3, can be used to investigate the CMV [4].

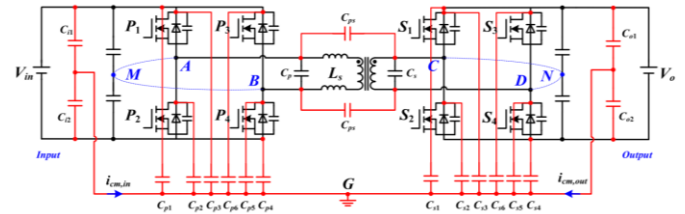


Figure 3 DAB circuit with parasitic capacitances[4]

3. Modulation methods

The main difference of the three modulation methods has been indicated in Figure 4, where V_p and V_s are representing the voltages on the transformer winding.

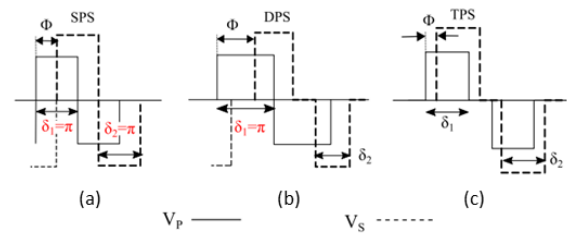


Figure 4 Phase-Shift Modulations (a) SPS (b) DPS (c) TPS [5]

3.1. SPS

SPS modulation is the easiest way to control the DAB. Both H-Bridges switch with a constant pulse-width-modulated with 50% duty cycle. As a result, high frequency square wave voltages V_1 and V_2 are generated with the amplitude of $\pm V_i$ (input voltage) and $\pm V_o$ (output voltage) in the primary and secondary transformer terminals respectively. To generate the phase shift in these voltages a phase shift (φ) must be considered between the same switches of the two bridges. Therefore, it's possible to generate a current in the inductor and transfer power from the leading side to the other one. The main waveforms of SPS modulation are shown in Figure 5 [3], [6].

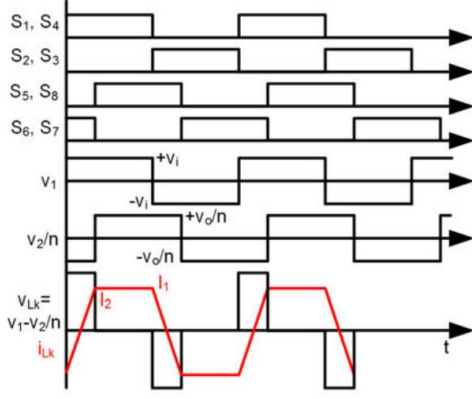


Figure 5 DAB waveforms with SPS modulation [6]

Using the equivalent circuit in Figure 2 and the Faraday law for the inductor, it's possible to calculate I_1 and I_2 , indicated in Figure 5, as:

$$I_1 = [(2D - 1) + M] \frac{V_i T_s}{4L} \quad \text{Eq. 2}$$

$$I_2 = [1 + M(2D - 1)] \frac{V_i T_s}{4L} \quad \text{Eq. 3}$$

where $D = \frac{\varphi T_s}{2}$ is the duty cycle, M is the voltage conversion ratio ($M = \frac{V_o}{nV_i}$) and L is the equivalent series inductance of the circuit. The total power that can be transferred can be calculated as:

$$P = \frac{MV_i^2 D(1 - D)T_s}{2L} \quad \text{Eq. 4}$$

The ZVS can be ensured if I_1 and I_2 are positive. If $M=1$, they are always positive. If $M>1$, I_1 is always positive and I_2 is positive if:

$$D > \frac{M - 1}{2M} \quad \text{Eq. 5}$$

On the other hand, if $M<1$, I_2 is always positive and I_1 is positive for

$$D > \frac{1 - M}{2} \quad \text{Eq. 6}$$

Therefore, it's possible to plot the ZVS region of DAB obtained is using SPS modulation as shown in Figure 6.

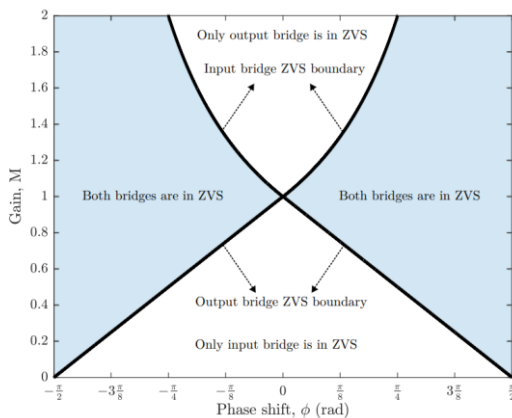


Figure 6 ZVS operating region in SPS modulation[3]

3.2. DPS

Dual phase-shift control is introduced to solve the limitations of SPS control. The basic idea in dual phase shift control is to add an extra degree of freedom by introducing the phase-shift between the pairs of the bridges (e.g., S1 and S4). In this way this phase-shift -which from now on we call it inner phase shift- will introduce one extra degree of freedom which can be employed to solve some problems. It's possible to reduce and eliminate the reactive power, reduce the peak of the inductor current to increase the efficiency or reduce the current stresses on the switches.

The inner (D1) and outer (D2) phase shifts define four possible working conditions as shown in Table 1.

Table 1 DPS operating cases [7]

Case	Conditions	
A	$D1 \geq D2$	$D1 + D2 \leq 1$
B	$D1 \geq D2$	$D1 + D2 > 1$
C	$D1 < D2$	$D1 + D2 \leq 1$
D	$D1 < D2$	$D1 + D2 > 1$

In this thesis, only the case B was analysed since it is the only case useful to decrease the RMS of the current in the DAB. The waveforms of the DAB using DPS are shown in Figure 7 along with the conducting switches in each time interval.

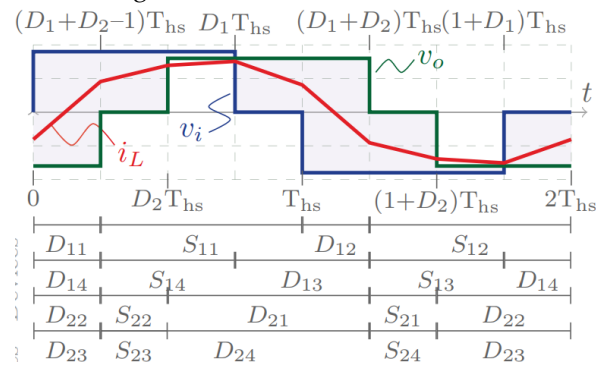


Figure 7 Ideal Voltage and currents in DPS modulation [7]

If all the powers and voltages transferred to the per unit space, considering the equivalent circuit of DAB, it's possible to calculate the currents and powers as in Table 2. where $k = \frac{nV_o}{V_i}$.

To find the ZVS range of DAB using DPS, the same principle should be taken into account. Using the waveforms, it's possible to find that for what values of phase shift ratios the current is negative

at the moment of switching. It can be calculated that for the DPS modulation to have ZVS the conditions in Eq. 7 and Eq. 8 should be respected.

Table 2 Voltages and currents formula in DPS[7]

I_o	$2[-(1-k)D_1^2 + (1-k)D_1 + 2kD_1D_2]$
I_{rms}	$\frac{2}{\sqrt{3}}\sqrt{(1-k)^2(3-2D_1)D_1^2 + 4k(3D_1-D_2)D_2^2}$
P	$2k[2D_1D_2 - D_2^2]$
S	$\frac{(1+k)}{\sqrt{3}}\sqrt{(1-k)^2(3-2D_1)D_1^3 + 4kD_1D_2^2(3D_1-D_2)}$

$$D_2 \geq 1 - \frac{1+k}{2k}D_1 \quad \text{Eq. 7}$$

$$D_2 \geq \frac{1-k}{2}D_1 \quad \text{Eq. 8}$$

There are numerous pairs of phase shifts which will transfer the same power. Therefore, a minimization algorithm has been used to find the best pairs of phase shifts which lead to minimum RMS of the current.

By getting a partial derivative from Lagrangian equation in Eq. 9, (λ is the lagrangian multiplier) it's possible to find the minimum S (apparent power) which leads to the least RMS of the current.

$$L(D_1, D_2, \lambda) = S(D_1, D_2) + \lambda[P(D_1, D_2) - P_{ref}] \quad \text{Eq. 9}$$

Finally, the pairs of phase shifts leading to the minimum RMS of the current is achieved from:

$$\begin{cases} (1-2D_2)(1-D_2)^3 + 3D_1(4D_2-D_1-1)(1-D_2)^2 \\ -6D_1^2(1-D_1)(1-D_2) - (1-2D_2)(D_2-D_1)^3 = 0 \\ -3D_1(2D_2+D_1-2)(D_2-D_1)^2 + D_1^2(1-2D_2)(9-8D_1) \\ 2[D_1(2-D_1) + 2D_2(1-D_2) - 1] - P_{ref} = 0 \end{cases} \quad \text{Eq. 10}$$

3.3. TPS

In TPS there are three degrees of freedom that can introduce more flexibility for controlling the DAB converter. The main waveform of the TPS modulation method is shown in Figure 8. The phase shifts are introduced with respect to S1 [8]. It is possible to demonstrate that, either D2 is smaller or larger than D3, the same amount of currents and voltages will be achieved and the same amount of power will be transferred. Thus, for the sake of simplicity, we assume that D2 is always lower than D3. Hence, it's possible to analyze the circuit in 3 different situations which are

$$\begin{cases} 0 \leq D_1 \leq D_2 \leq D_3 \leq 1 \\ 0 \leq D_2 \leq D_1 \leq D_3 \leq 1 \\ 0 \leq D_2 \leq D_3 \leq D_1 \leq 1 \end{cases} \quad \text{Eq. 11}$$

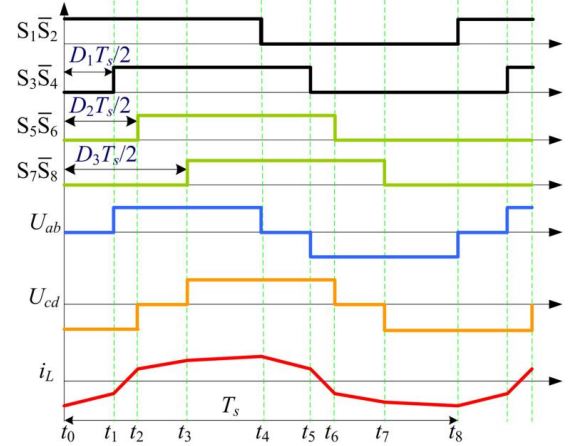


Figure 8 Waveforms of DAB using TPS modulation [8]

The algorithm which was used to maintain the phase shifts was aimed to reduce the current stresses on the switches. For $k > 1$, it's possible to calculate the unified current stress (i_p) as:

$$i_p = \frac{i_p}{i_N} = 2[D_2 + D_3 - 1 + k(1 - D_1)] \quad \text{Eq. 12}$$

Moreover, by calculating the voltage across the inductor, the unified output power can be written as:

$$P = \frac{P}{P_N} = \begin{cases} 2 \left(\frac{-D_1 + D_2 + D_3 - D_1^2}{-D_2^2 - D_3^2 + D_1D_2 + D_1D_3} \right) & D_1 \leq D_2 \leq D_3 \leq 1 \\ 2 \left(\frac{-D_1 + D_2 + D_3}{+D_1D_3 - D_3^2 - D_1D_2} \right) & D_2 \leq D_1 \leq D_3 \leq 1 \\ 2 \left(\frac{-D_1 + D_2 + D_3}{+D_1^2 - D_1D_2 - D_1D_3} \right) & D_2 \leq D_3 \leq D_1 \leq 1 \end{cases} \quad \text{Eq. 13}$$

where P_N is:

$$P_N = \frac{nT_s U_{in} U_o}{8L} \quad \text{Eq. 14}$$

By applying partial derivative in the Lagrangian function in Eq. 15 and solving the equations equal to zero, it's possible to find formulas for obtaining the group of phase shifts leading to the minimum current stress as in Eq. 16 and Eq. 17.

$$E = i_p + \lambda(P - P^*) \quad \text{Eq. 15}$$

$$\begin{cases} D_2 = \begin{cases} (k-1)(1-D_1) & 0 \leq p < \frac{2(k-1)}{k^2} \\ \frac{k-2}{2(k-1)}D_1 + \frac{1}{2} & \frac{2(k-1)}{k^2} < p \leq 1 \end{cases} \\ D_3 = \begin{cases} D_1 & 0 \leq p < \frac{2(k-1)}{k^2} \\ \frac{k-2}{2(k-1)}D_1 + \frac{1}{2} & \frac{2(k-1)}{k^2} < p \leq 1 \end{cases} \end{cases} \quad \text{Eq. 16}$$

$$D_1 = \begin{cases} 1 - \frac{\sqrt{2p(k-1)}}{2k-2} & 0 \leq p < \frac{2(k-1)}{k^2} \\ \frac{(k-1)\sqrt{(1-p)(k^2-2k+2)}}{k^2+2k+2} & \frac{2(k-1)}{k^2} < p \leq 1 \end{cases} \quad \text{Eq. 17}$$

4. Simulation results

The information of the case study that was used in the simulations have been summarised in Table 3.

Table 3 Design parameters

Parameter	Value
Input Voltage	400 V
Nominal Output Voltage	800 V
Nominal Power	10 kW
Switching Frequency	10 kHz
Load	64 Ω
Inductance	18.2 μH
Output capacitance	750 μF
Transformer turn ratio	1:2

Based on the calculation, “NTMT064N65S3H” mosfet was chosen for the input bridge and “FCH085N80” was chosen for the output bridge. In order to validate the results obtained by the Simulink, a model in Ltspice was created using the spice model of the MOSFETs provided by the ONSEMI company. The validation was done for 10 kW and 6 kW. After validating the Simulink results was reasonable with a good tolerance of 0.8% at maximum. In other words, in the worst case, the efficiency in Simulink was equal to 99.2% of efficiency in Ltspice. Therefore, all the Simulink results can be reliable.

In order to prevent the effect of dead time in all the modulations, a compensation was made in phase shifts between the bridges. In this way the voltages had the required phase shift for every power without any voltage sag or change in the polarity. The efficiency of the DAB with the three modulations is shown in Figure 9. It is obvious that the efficiency is high, using TPS modulation specially in low powers. The reason of this big difference is that the TPS modulation is fully ZVS, but the other two modulations will lose ZVS around 4 kW in DPS and 5 kW in SPS, as it is shown in Figure 10. For both DPS and SPS modulation, the ZVS was lost in the HV side bridge.

SPS has the highest efficiency of all, within 4.5 kW to around 7 kW. An important point here is that in the whole range of 4-10 kW, the DPS has the lowest.

In both DPS and TPS modulation, the goal of optimization was to lower the RMS current of the input. Figure 11 shows that both methods were successfully able to reduce the RMS current with

respect to SPS modulation. At low powers this reduction is much more in TPS method which is the reason why the conduction losses are much lower in this method at lower power. At very high power, close to the nominal one, the DPS modulation has slightly a better performance than TPS.

Concerning the common mode voltages, the following statements can summarize the difference of CMV in each modulation.

- In SPS, since the bridges' voltages are similar by means of phase shifts, the CMV in input and output are the same. But in TPS and DPS the input and output CMV are different from each other judging by the ripples.
- In SPS, the MOSFETs parasitic capacitances does not have any effect on the other side, this is true also for TPS in high powers. In contrast, in low powers using TPS it has proven that the LV side MOSFETs also influence the HV side CMV. In DPS there are some effects such as displacements and ripple reduction in the other bridge.
- The ripples in SPS are simpler and are equal always. But in DPS and TPS the value of ripples is different in one period. It can be seen that because of the phase shifts the ripples values can be different. It should be considered that a same phenomenon has the same ripple value in every modulation. What really happens is that the effect of phase shifts cause ripples to sum or subtract with each other which leads to different value of ripples for same values of voltages.

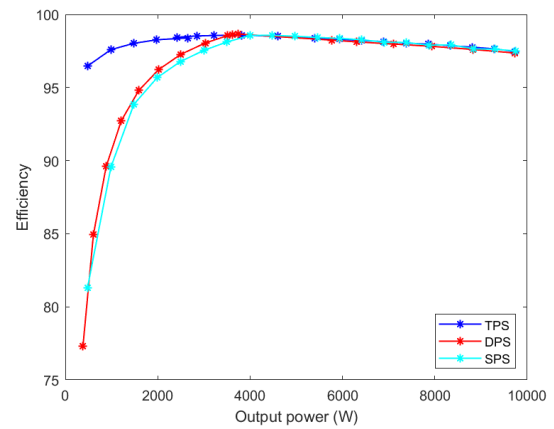


Figure 9 : Efficiency of different modulations

5. Conclusion

The three modulation methods have been compared with each other based on the simulations' results. A summary of all the results and experiments has been reported in Table 4.

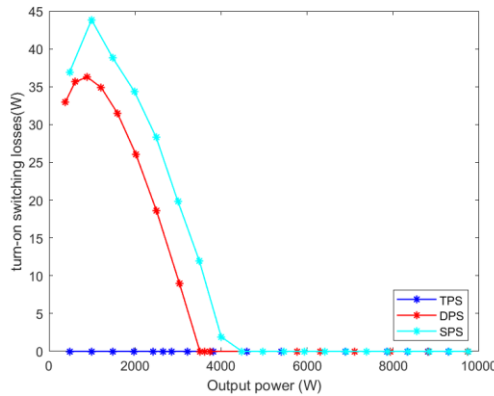


Figure 10 ZVS range in different modulations

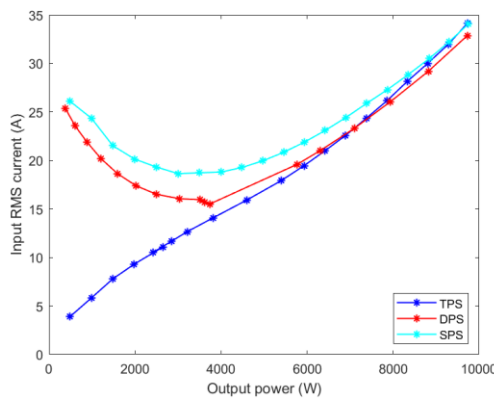


Figure 11 Input RMS current

Table 4 Summary of comparison

	SPS	DPS	TPS
Complexity of modulation	Low	Medium	High
Number of phase shifts	1	2	3
Usage of algorithm	No	Yes	Yes
Efficiency at high power	High	Low	High
Efficiency at low power	Lowest	medium	Highest
variation of efficiency	81.28% ~ 98.57%	77.03%~ 98.66%	96.49% ~ 98.59%
ZVS range	3.7 kW ~ 10 kW	4.5 kW ~ 10 kW	Full range
Dead time compensation	Applied in HV side	Applied in HV side	Applied in LV side
ZVS-off in HV bridge (in low powers)	Yes	Yes	No
Conduction losses	Medium	Highest	Lowest
Input RMS current	Highest	Medium	Lowest
CMV Ripple	Medium	Highest	Medium

All in all, based on this thesis it's possible to say that if the whole range of power is required for the application the TPS modulation will be a good choice since it has the full range ZVS. Otherwise, at high power, close to the rated one, the efficiency of SPS modulation is very close to TPS. Since the SPS is much simpler to implement, if only the high power range is required, the SPS modulation would be a good choice. Also, from the CMV point of view, the SPS has much less ripples and smaller values.

The DPS modulation, used in this thesis, was not able to perform good neither for efficiency in high or low powers nor for the CMV. Therefore, between these three the author would not suggest the DPS to be used.

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